

MrcPunc: Modulation and Rate Compatible Puncturing for Few-Iteration 5G LDPC Decoding

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Abstract—To meet the stringent high-throughput demands of enhanced mobile broadband (eMBB), low-density parity-check (LDPC) decoding is limited to few iterations. This constraint usually degrades error-correcting performance, rousing a critical need for puncturing towards algorithmic enhancements. However, existing puncturing methods lack modulation and rate compatibility, converging to local optima due to greedy search strategies. To this end, this paper proposes a modulation and rate compatible puncturing (MrcPunc) for 5G LDPC codes. It incorporates three key techniques: 1) a revised protograph-based extrinsic information transfer (PEXIT) analysis for accurate performance prediction, 2) a progressive multi-path search (PMPS) algorithm to avoid local optima, and 3) a compatibility optimization supporting various modulations and code rates. The MrcPunc yields up to 0.6 dB gain over the 5G standard puncturing under few-iteration condition without extra decoding complexity. It is noted that MrcPunc guarantees robustness against both larger iteration counts and mismatched modulations, alongside reducing the average number of decoding iterations.

Index Terms—Puncturing, compatibility, low-density parity-check codes, few-iteration decoding.

I. INTRODUCTION

SCENARIOS of the fifth-generation (5G) mobile networks, such as ultra-reliable low-latency communications (URLLC) and enhanced mobile broadband (eMBB), impose more stringent requirements, including sub-millisecond latency (≤ 0.1 ms) and peak data rates exceeding 1 Tbps [1]. The large-iteration 5G low-density parity-check (LDPC) decoding poses challenges in meeting these requirements, as

the decoding latency increases linearly with iteration counts. Moreover, both URLLC and eMBB require 5G LDPC codes to maintain robust error-correcting capabilities [2], [3], [4], [5]. Although simplifying decoding and refining codes can mitigate issues related to latency, throughput, and reliability, respectively, a reduction in iterations inevitably leads to significant performance degradation.

To address the trade-off between low-latency, high-throughput, and ultra-reliability, a series of low-complexity LDPC decoding have been proposed, maintaining comparable performance via parameter optimization [6], [7], [8], [9]. However, these algorithms are inherently limited by the performance bounds of belief-propagation (BP) decoding, making LDPC design a pivotal alternative to enhance error-correcting capabilities through the optimization of code ensembles. Conventional LDPC design relies on density evolution (DE) analysis for evaluation [10]. Chung et al. optimized an LDPC ensemble using discrete DE, achieving near-Shannon-limit performance but requiring extensive iterations [11]. The studies, such as [11], [12], [13], and [14], assume near-infinite iterations, focusing only on large-iteration decoding and resulting in the suboptimal few-iteration decoding.

The concept of few-iteration decoding was first introduced in [15], which highlighted shortcomings of the 5G standard puncturing. Through direct simulations, the authors optimized the puncturing for moderate-rate and short 5G LDPC codes with 5 $\sim$ 10 iterations, achieving performance gains of up to 0.55 dB. Similarly, Koike-Akino et al. [16], [17] concentrated on iteration-aware decoding, developing LDPC codes with better performance compared to conventional ones by leveraging extrinsic information transfer (EXIT) analysis, an approximation of DE. However, these approaches present inherent limitations: the simulation-based optimization in [15] encounters a significant computational bottleneck due to the vast combinatorial space involving code rates, lengths, and modulations; although the EXIT analysis in [16] and [17] accelerates performance evaluation, it is confined to the LDPC design for specific code rates and low-order modulations. This lack of modulation and rate compatibility severely limits practical applicability in the future evolution of 5G new radio (NR) systems, where adaptive code rates and modulations are essential to accommodate dynamically varying channel conditions.

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In contrast to [15], the DE-like analysis is utilized for puncturing optimization across different channels and decoding in [18], [19], [20], [21], and [22], while a range of rate compatible puncturing methods have been introduced in [23], [24], [25], [26], and [27]. These methods focus solely on the node perspective and neglect the protograph-based characteristic of 5G LDPC codes, similar to EXIT analysis. Although the protograph-based EXIT (PEXIT) analysis [28] acknowledges this aspect, it fails to account for the influence of few-iteration decoding and overlooks differences in information propagation paths between flooding and layered decoding. This oversight results in puncturing with poor performance being evaluated as optimal. Consequently, optimizations employing the traditional PEXIT analysis fail to achieve optimal puncturing, particularly in the context of layered few-iteration decoding, which is inevitable in URLLC and eMBB scenarios with more stringent requirements. Motivated by these considerations, this paper seeks to utilize the revised PEXIT analysis to enhance the efficiency of puncturing evaluation, while ensuring that the puncturing aligns with modulation and rate compatibility requirements, thereby addressing the challenges of adaptability in 5G and beyond 5G systems.

The main contributions of this work are summarized as follows.

- As the introduction of a layered schedule alters information propagation paths, the traditional PEXIT analysis fails to evaluate the puncturing for layered few-iteration decoding. Thus, we develop a revised PEXIT analysis for puncturing evaluation in bit-interleaved coded modulation (BICM) systems.
- To address the local optimum issue of the greedy search strategies in [15], we propose a progressive multi-path search (PMPS) algorithm to enhance the puncturing optimization, especially under high-order modulations, while ensuring acceptable computational complexity.
- To meet the compatibility requirement of 5G NR systems, we develop modulation and rate compatible puncturing (MrcPunc) for 5G LDPC decoding with different iteration counts, significantly enhancing error-correcting performance across different code rates, lengths, and modulations.
- We propose a block interleaver to implement MrcPunc, which avoids significant modifications to the existing 5G NR system and enables MrcPunc to be directly integrated with incremental redundancy hybrid automatic repeat request (IR-HARQ) transmission.

The remainder of this paper is organized as follows: Section II overviews the characteristics of 5G LDPC codes and puncturing, modulation and coding scheme (MCS) tables, as well as bit-interleavers in 5G NR systems. Section III develops the layered PEXIT analysis and proves its validity. Section IV details the design of MrcPunc. Section V presents the implementation of MrcPunc and its combination with IR-HARQ transmission. Section VI thoroughly analyzes the advantages of the PMPS algorithm and MrcPunc. Finally, Section VII concludes this paper with key findings and insights.

Notations: Boldface letters are used to represent vectors or matrices, where for a vector $\mathbf{a} = [a_i]_{1 \leq i \leq n}$, a_i is the i -th

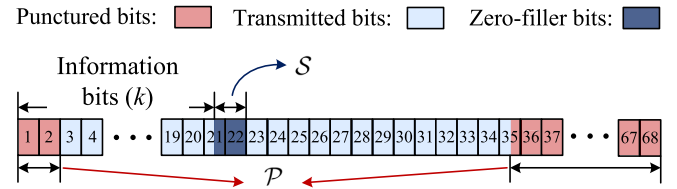


Fig. 1. The 5G standard puncturing for the BG1.

element of the vector $\mathbf{a}$, and for a matrix $\mathbf{b} = [b_{i,j}]_{1 \leq i \leq m, 1 \leq j \leq n}$, $b_{i,j}$ is the element at the i -th row and j -th column of the matrix $\mathbf{b}$. The notation $\|\mathbf{a}\|_1$ denotes the L1 norm of the vector $\mathbf{a}$, which satisfies $\|\mathbf{a}\|_1 = \sum_{i=1}^n a_i$. $\mathbf{e}_i$ denotes a vector where the i -th element is 1 and all other elements are 0. Calligraphic letters such as $\mathcal{A}$ represent sets, with $|\mathcal{A}|$ denoting the number of elements in the set $\mathcal{A}$. $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ are the floor function and ceiling function respectively. The complementary error function, denoted as $\text{erfc}(\cdot)$, is expressed by $\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-y^2} dy$.

II. PRELIMINARIES

In this section, we introduce the parity-check matrices of 5G LDPC codes, the 5G standard puncturing, the characteristics of BICM systems, and the MCS tables.

A. LDPC Codes and Puncturing in 5G Standard

5G LDPC codes are a class of protograph-based LDPC codes [29], [30], where the parity-check matrices all can be structured as an array of $Z \times Z$ submatrices, with Z denoting the lifting factor. The base matrix $\mathbf{H}_b = [b_{h,s}]_{1 \leq h \leq m, 1 \leq s \leq n}$ specifies the array structure, where $b_{h,s} \in \mathbb{N}$ corresponds to a $Z \times Z$ permutation matrix or all-zero matrix, m and n represent the numbers of rows and columns in the base matrix respectively. Thus, the Tanner graph of a 5G LDPC code consists of $m \times Z$ check nodes (CNs) and $n \times Z$ variable nodes (VNs), where the i -th CN and j -th VN are denoted as c_i and v_j , respectively. According to these characteristics of 5G LDPC codes, CNs and VNs can be grouped into CN blocks and VN blocks, each containing Z nodes. The base matrices of 5G LDPC codes are derived from the base graph (BG) 1 and BG2, enabling the construction of codes with flexible code lengths and a wide range of code rates.

For information bits of length k , if $k \notin \{iZ \mid i \in \mathbb{N}\}$, zero-filler is applied to the end of these bits before channel encoding, and the set of filler VNs is denoted as $\mathcal{S}$. These filler bits are not transmitted since they are known a priori at the receiver ends. During decoding, these VNs are either assigned the maximum log-likelihood ratio (LLR) or completely removed from the Tanner graph. After encoding, 5G NR systems perform puncturing on VNs to achieve the target code rate R . As illustrated in Fig. 1, the 5G standard prioritizes the first two VN blocks for puncturing, followed by sequential puncturing from the last VN backward. The set of punctured VNs, defined as $\mathcal{P}$, corresponds to the bits that are not transmitted and unknown at the receiver. After zero-filling and puncturing, the communication system can achieve

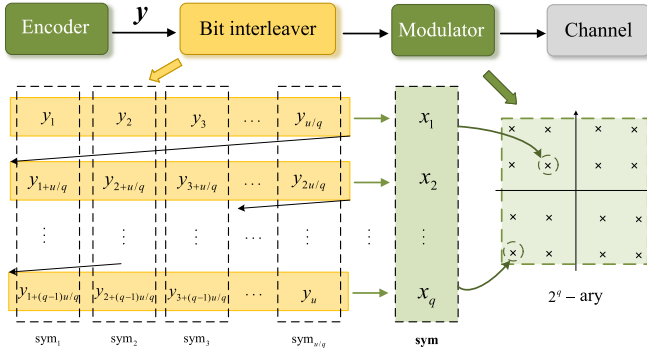


Fig. 2. A BICM system and the mode- q row-column interleaver.

the target code rate $R = \frac{k}{nZ - |\mathcal{S}| - |\mathcal{P}|}$. For the first two and partially punctured VN blocks, the corresponding input LLRs are set to zero for the decoders. The 5G standard puncturing is optimal in large-iteration decoding but becomes suboptimal in few-iteration decoding [15], as it is primarily designed for the latter [31].

B. BICM and MCS Tables in 5G NR Systems

In the 5G NR systems, a Gray-mapping square uniform 2^q -quadrature amplitude modulation (QAM) is employed to improve spectral efficiency, where $q \in \{2, 4, 6, 8, 10\}$. When $q = 2$, the modulation reduces to the quadrature phase shift keying (QPSK). The transmission sequence, denoted as $\mathbf{y} = [y_h]_{1 \leq h \leq u}$, originates from the information bits encoded by the encoder, and zero-filled to length u before bit interleaving, where $u = q \lceil (nZ - |\mathcal{S}| - |\mathcal{P}|) / q \rceil$. As shown in Fig. 2, the encoder, a mode- q row-column interleaver, and modulator compose a BICM system, and the interleaver maps the leading bits of the 5G LDPC codeword to the leading bits of modulated symbols via row-wise writing and column-wise reading. Each symbol in 2^q -QAM is represented as a vector $\mathbf{x} = [x_h]_{1 \leq h \leq q}$, where the error probability of the h -th bit x_h provided in [32] satisfies

$$P_e^{(h)}(\sigma) = \frac{1}{2^{q/2}} \sum_{s=0}^{(1-2^{-h'})2^{q/2}-1} \left[(-1)^{\left\lfloor \frac{s \cdot 2^{h'-1}}{2^{q/2}} \right\rfloor} \times \left(2^{h'-1} - \left\lfloor \frac{s \cdot 2^{h'-1}}{2^{q/2}} + \frac{1}{2} \right\rfloor \right) \times \operatorname{erfc} \left((2s+1) \sqrt{\frac{3}{(2^q-1)\sigma^2}} \right) \right], \quad (1)$$

in an additive white Gaussian noise (AWGN) channel. The σ^2 represents the sum of the noise variance for the in-phase and quadrature channels, $h' = \lceil h/2 \rceil$, and $P_e^{(h)}(\sigma)$ also satisfies $P_e^{(1)}(\sigma) = P_e^{(2)}(\sigma) < P_e^{(3)}(\sigma) = P_e^{(4)}(\sigma) < \dots < P_e^{(q-1)}(\sigma) = P_e^{(q)}(\sigma)$. Combining the error probability ordering of constellation label bits with the VN degree distribution of 5G LDPC codes, the row-column interleaver used in the 5G standard approximates a variable degree matched mapping interleaver, which typically leads to a significant performance gain under high-order modulations [33], [34], [35].

TABLE I
THE MCS TABLE FOR 5G LDPC CODES FROM THE BG1

R (Code Rate)	Modulation
1/3 ~ 11/26	QPSK, 16-QAM
22/51 ~ 11/17	QPSK, 16-QAM, 64-QAM
2/3 ~ 11/16	QPSK, 16-QAM, 64-QAM, 256-QAM
22/31 ~ 22/29	64-QAM, 256-QAM
11/14 ~ 11/12	64-QAM, 256-QAM, 1024-QAM

In the 5G physical downlink shared channel, the communication system adapts the code rate and modulation according to channel quality to balance reliability and spectral efficiency—employing high code rates and high-order modulation in high-quality channels, and low code rates and low-order modulations otherwise. Strict correspondences exist between different code rates and modulations, described by four MCS tables in the 5G standard [29]. When $|\mathcal{S}| = 0$ and no VN blocks are partially punctured ($|\mathcal{P}| = hZ, h \in \mathbb{N}$), the MCS table for 5G LDPC codes from BG1 is shown in Table I, which illustrates a specific matching relationship between the different code rates and modulations.

III. LAYERED PEXIT ANALYSIS FOR BICM SYSTEMS

In this section, a layered PEXIT analysis is proposed to assess the performance of the layered few-iteration decoding. When the iteration counts are small, the traditional PEXIT analysis [28], [36] cannot capture the convergence of layered decoding. Furthermore, changes in the decoding neighborhood induced by layered scheduling are considered. Before deriving the layered PEXIT analysis, we prove the validity of the fundamental assumption in DE analysis: the decoding neighborhood is a tree for infinite-length codes.

A. Flooding and Layered Decoding

The conventional decoding of LDPC codes employs BP algorithm [37], which iteratively exchanges messages between VNs and CNs to update LLRs of bits in the codeword. Demodulated and zero-filled noisy channel information, denoted as $l_j^{(0)}$, is directly fed into the VN v_j . During the t -th iteration, each VN passes information to all connected CNs (the edge information from v_j to c_i is denoted as $l_{j \rightarrow i}^{(t)}$), while each CN updates and passes information to all connected VN (the message from c_i to v_j is denoted as $l_{i \rightarrow j}^{(t)}$). After each iteration, the updated LLR, denoted as $l_j^{(t)}$, is output by the VN v_j to check whether to early terminate the decoding process. The flooding BP (FBP) decoding is updated iteratively through flooding scheduling. The layered BP decoding (LBP) updates a subset of CNs layer by layer in each iteration to achieve a faster decoding convergence speed and better performance [38], [39].

Consider a Tanner graph for a protograph-based LDPC code with mZ CNs $\{c_1, \dots, c_{mZ}\}$ and nZ VNs $\{v_1, \dots, v_{nZ}\}$. All CNs are divided into m' sets $\mathcal{C}_1, \dots, \mathcal{C}_{m'}$ such that $\bigcup_{i=1}^{m'} \mathcal{C}_i = \{c_1, c_2, \dots, c_{mZ}\}$ and $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$ for $i \neq j$. In one iteration of

the LBP decoding, information updates are performed from the first layer to the m' -th layer.¹ The i -th layer only updates the information passed by the CNs in C_i and the VNs connected to them in the Tanner graph. In this paper, we assume that each C_i contains exactly the Z CNs expanded from a single CN block in the protograph. In contrast, the FBP decoding consists of only one layer, and the corresponding set C_1 of this layer contains all CNs.

B. Layered Decoding Neighborhood

The layered decoding neighborhood $\mathcal{L}^{(t_M)}(v_j)$ is a tree-like graph rooted at v_j . $\mathcal{L}^{(t_M)}(v_j)$ describes the dependency structure contributing to the decoding of v_j within t_M layered iterations. Notably, if the neighborhood contains repeated nodes, these duplicates are treated as distinct entities. When referring to a node o in the neighborhood, we specifically mean the instance in the dependency path, not all its possible duplicates. In a layered decoding neighborhood, let $\mathcal{N}(o)$ be the neighbor set of o , which contains the nodes connected to o in the graph. $\tilde{\mathcal{N}}(o) \subset \mathcal{N}(o)$ will be guaranteed, where $\mathcal{N}(o)$ is the neighbor set of o in the Tanner graph. The rigorous definition of $\mathcal{L}^{(t_M)}(v_j)$ is given as follows.

Definition 1: The layered decoding neighborhood $\mathcal{L}^{(t_M)}(v_j)$ is constructed recursively using an index $\kappa = 0, 1, \dots, m, t_M$, where each κ corresponds to a distinct layer within one iteration. The temporary neighborhood at step κ is denoted by $\mathcal{L}^{(t_M)}(v_j; \kappa)$, and the final neighborhood is $\mathcal{L}^{(t_M)}(v_j) = \mathcal{L}^{(t_M)}(v_j; mt_M)$. Prior to the recursive expansion, the temporary neighborhood $\mathcal{L}^{(t_M)}(v_j; 0)$ is initialized to consist solely of the root node v_j . For each recursive expansion step $\kappa \geq 1$, $\mathcal{L}^{(t_M)}(v_j; \kappa - 1)$ is a subtree of $\mathcal{L}^{(t_M)}(v_j; \kappa)$, and the index of CN block to be updated in this layer is $m - \ell(\kappa) + 1$, where $\ell(\kappa) = [(\kappa - 1) \bmod m] + 1$, and we perform as follows:

1) *Add CNs to the neighborhood:* For each VN $v_{j'}$ in $\mathcal{L}^{(t_M)}(v_j; \kappa - 1)$ with $|\tilde{\mathcal{N}}(v_{j'})| < |\mathcal{N}(v_{j'})|$, connect the CNs in $(\mathcal{N}(v_{j'}) \cap C_{m-\ell(\kappa)+1}) \setminus \tilde{\mathcal{N}}(v_{j'})$ to $v_{j'}$.

2) *Add VNs to the neighborhood:* For each newly added CN c_i , connect the VNs in $\mathcal{N}(c_i) \setminus \tilde{\mathcal{N}}(c_i)$ to c_i .

Remark 1: In the above definition, the index $\kappa \geq 1$ has a practical meaning related to layered decoding. Let $t(\kappa) = \lceil \kappa/m \rceil$. The newly added VNs and CNs are nodes updated in the $(m - \ell(\kappa) + 1)$ -th layer of the $(t_M - t(\kappa) + 1)$ -th iteration.

Example 1: Consider a base graph matrix of a protograph-based LDPC code:

$$\mathbf{H}_b = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}. \quad (2)$$

When $Z = 3$, the lifted parity-check matrix is given as

$$\mathbf{H} = \begin{bmatrix} 1 & & & & 1 & & & 1 \\ & 1 & & & & 1 & & 1 \\ & & \mathbf{O} & & & & & \\ & & & 1 & & & & \\ 1 & & & & 1 & & 1 & \\ & 1 & & & & 1 & & \\ & & 1 & & & & & \mathbf{O} \\ & & & 1 & & & & \\ & & & & 1 & & & \end{bmatrix}. \quad (3)$$

¹The schedule of different layers in the decoding is also an aspect of the optimization. However, this paper does not consider this and assumes sequential decoding from the first layer to the m' -th layer by default.

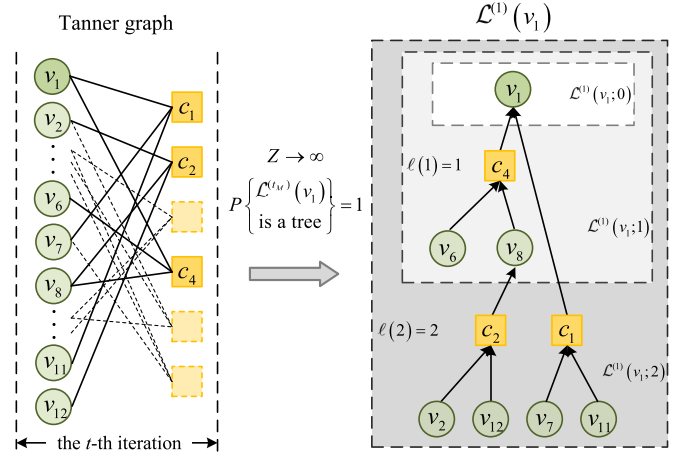


Fig. 3. An example of the layered decoding neighborhood $\mathcal{L}^{(1)}(v_1)$ for one iteration obtained from the Tanner graph.

Fig. 3 illustrates an example of the layered decoding neighborhood $\mathcal{L}^{(1)}(v_1)$ for one iteration obtained from the Tanner graph. $\mathcal{L}^{(1)}(v_1; 0)$ only contains the root node v_1 where the final information converges. The construction of $\mathcal{L}^{(1)}(v_1; 1)$ is divided into two steps according to the rules of Definition 1. The first step is adding the CNs in the second decoding layer. Since $\mathcal{N}(v_1) \cap C_2 \setminus \tilde{\mathcal{N}}(v_1) = \{c_4\}$, thus only c_4 is connected to v_1 . The second step is adding VNs which should be connected to the newly added CNs. Since $\mathcal{N}(c_4) \setminus \tilde{\mathcal{N}}(c_4) = \{v_6, v_8\}$, thus the newly added v_6 and v_8 are connected to c_4 . Similarly, $\mathcal{L}^{(1)}(v_1; 2)$ is constructed based on $\mathcal{L}^{(1)}(v_1; 1)$, where all newly added CNs are in the set $\mathcal{N}(\{v_1, v_6, v_8\}) \cap C_1 \setminus \tilde{\mathcal{N}}(\{v_1, v_6, v_8\}) = \{c_1, c_2\}$, and all newly added VNs are in the set $\mathcal{N}(\{c_1, c_2\}) \setminus \tilde{\mathcal{N}}(\{c_1, c_2\}) = \{v_2, v_7, v_{11}, v_{12}\}$. When it is necessary to continue constructing the layered decoding neighborhood corresponding to a larger iteration count, new nodes will be recursively added on the basis of $\mathcal{L}^{(1)}(v_1; 2)$ in accordance with Definition 1.

C. Derivation of Layered PEXIT Analysis

We aim to develop a layered PEXIT analysis, which overcomes the mismatch between traditional PEXIT analysis and layered decoding. The PEXIT analysis evaluates decoding performance especially for punctured codes by virtue of accounting for differences and connection specifics between VN and CN blocks [28], [36]. This approach determines whether the decoder can converge at a certain signal-to-noise ratio (SNR) by calculating the mutual information, which is denoted by $I_{APP,s}^{(t)}$, between the soft outputs of each VN block and the transmitted codewords in the t -th iteration. The mutual information between the edge information from the s -th VN block to the h -th CN block (or vice versa) and the transmitted codewords in the t -th iteration is define as $I_{VN,s \rightarrow h}^{(t)}$ (and $I_{CN,h \rightarrow s}^{(t)}$), where the s -th VN block and the h -th CN block are the sets $\mathcal{V}_s = \{v_{(s-1)Z+1}, \dots, v_{sZ}\}$ and $\mathcal{C}_h = \{c_{(h-1)Z+1}, \dots, c_{hZ}\}$ respectively.

The following theorem indicates that for the protograph-based LDPC codes, when the lifting factor satisfies $Z \rightarrow \infty$, a layered decoding neighborhood converges to a tree, and then

the layered PEXIT analysis can be conducted based on such trees.

Theorem 1: When $Z \rightarrow \infty$, for a given base graph and arbitrary v_j , the probability that the layered decoding neighborhood $\mathcal{L}^{(t_M)}(v_j)$ is a tree approaches 1.

Proof: When no node repetition occurs in the layered decoding neighborhood $\mathcal{L}^{(t_M)}(v_j)$, the path is guaranteed to be loop-free. For $\kappa = 0$, $\mathcal{L}^{(t_M)}(v_j; 0)$ contains only the root node v_j , and obviously $\mathcal{L}^{(t_M)}(v_j; 0)$ is without loops. Assume that $\mathcal{L}^{(t_M)}(v_j; \kappa')$ is a tree, let $C(h, s; \kappa')$ ($V(h, s; \kappa')$) denote the number of all CNs (VNs) that contain in $\mathcal{C}_h \cap \mathcal{N}(\mathcal{V}_s)$ ($\mathcal{V}_s \cap \mathcal{N}(\mathcal{C}_h)$) and $\mathcal{L}^{(t_M)}(v_j; \kappa')$. For $\kappa = \kappa' + 1$, $\mathcal{L}^{(t_M)}(v_j; \kappa)$ is constructed by adding new nodes based on $\mathcal{L}^{(t_M)}(v_j; \kappa')$. When the edge connecting $v_{j'} \in \mathcal{V}_s$ and $c_i \in \mathcal{C}_{m-\ell(\kappa'+1)+1}$ is added, the probability that CNs do not repeat is

$$\begin{aligned} & \Pr \{\text{CNs do not repeat in the step } \kappa\} \\ &= \frac{b_{h^\dagger, s} [Z - \sum_{s'=1}^n (C(h^\dagger, s'; \kappa') - K(h^\dagger, s'))]}{b_{h^\dagger, s} \cdot Z - C(h^\dagger, s; \kappa') - K(h^\dagger, s)}, \end{aligned} \quad (4)$$

where $K(h, s)$ is the number of already added edges which connect $v'_j \in \mathcal{V}_s$ and $c_i \in \mathcal{C}_h$, and $h^\dagger = m - \ell(\kappa' + 1) + 1$. When the newly added edge exists, $b_{h^\dagger, s} > 0$ is hold, and then Eq. (4) can be bounded as

$$\begin{aligned} & \Pr \{\text{CNs do not repeat in the step } \kappa\} \\ & \geq \frac{b_{h^\dagger, s} [Z - \sum_{s'=1}^n (C(h^\dagger, s'; \kappa') - K(h^\dagger, s'))]}{b_{h^\dagger, s} \cdot Z} \\ & \geq 1 - \frac{\sum_{s'=1}^n (C(h^\dagger, s'; \kappa') - K(h^\dagger, s'))}{Z} \\ & \geq 1 - \frac{\sum_{s'=1}^n C(h^\dagger, s'; \kappa')}{Z}. \end{aligned} \quad (5)$$

Similarly, let $\tilde{K}(h, s)$ denote the number of already added edges which connect $v'_j \in \mathcal{V}_s$ and $c_i \in \mathcal{C}_h$ when adding new VNs, and then the probability that VNs do not repeat is

$$\begin{aligned} & \Pr \{\text{VNs do not repeat in the step } \kappa\} \\ &= \frac{b_{h^\dagger, s} [Z - \sum_{h'=1}^m (V(h', s; \kappa') - K(h', s))]}{b_{h^\dagger, s} \cdot Z - V(h^\dagger, s; \kappa') - K(h^\dagger, s)} \\ & \geq \frac{b_{h^\dagger, s} [Z - \sum_{h'=1}^m (V(h', s; \kappa') - K(h', s))]}{b_{h^\dagger, s} \cdot Z} \\ & \geq 1 - \frac{\sum_{h'=1}^m (V(h', s; \kappa') - K(h', s))}{Z} \\ & \geq 1 - \frac{\max_{s \in \{1, \dots, n\}} (\sum_{h'=1}^m V(h', s; \kappa'))}{Z}. \end{aligned} \quad (6)$$

Eqs. (5) and (6) are both bounded by variables that depend solely on κ' and are independent of the newly added edge. Thus, the probability that no loop exists in $\mathcal{L}^{(t_M)}(v_j; \kappa)$ is bounded as

$$\begin{aligned} & \Pr \{\text{no loop exists in the step } \kappa\} \\ & \geq \left(1 - \frac{\max_{s \in \{1, \dots, n\}} (\sum_{h'=1}^m V(h', s; \kappa'))}{Z} \right)^{N_v(\kappa) - N_v(\kappa')} \\ & \times \left(1 - \frac{\sum_{s'=1}^n C(h^\dagger, s'; \kappa')}{Z} \right)^{N_c(\kappa) - N_c(\kappa')}, \end{aligned} \quad (7)$$

where $N_c(\kappa)$ and $N_v(\kappa)$ denote the numbers of CNs and VNs in $\mathcal{L}^{(t_M)}(v_j; \kappa)$, respectively. Since the values of $C(h, s; \kappa)$, $V(h, s; \kappa)$, $N_c(\kappa)$, $N_v(\kappa)$ depend only on $\mathbf{H}_b$ and κ rather than Z , they can be treated as finite constants. When $Z \rightarrow \infty$, we have

$$\Pr \{\text{no loop exists in the step } \kappa\} = 1 - O(Z^{-1}). \quad (8)$$

It follows that

$$\Pr \{\mathcal{L}^{(t_M)}(v_j) \text{ is a tree}\} = (1 - O(Z^{-1}))^{mt_M - \kappa'} \quad (9)$$

converges to 1 for all finite m and t_M . ■

Theorem 1 states that for a protograph-based LDPC code, its layered decoding neighborhood can be treated as a tree for decoding performance analysis when $Z \rightarrow \infty$. If the decoding neighborhood is not a tree, it can be approximated as a tree to characterize the asymptotic decoding performance [40], [41]. For a finite Z , the layered decoding neighborhood has the probability of being a tree as given in Eq. (9), and can also be approximated as a tree for decoding performance analysis. First, the mutual information passed from $v_j \in \tilde{\mathcal{N}}(c_i)$ to c_i is updated by

$$I(l_{j \rightarrow i}^{(t)}) \leftarrow \Phi \left[\Phi^{-1} \left(I(l_j^{(t)}) \right) - \Phi^{-1} \left(I(l_{i \rightarrow j}^{(t-1)}) \right) \right], \quad (10)$$

where $I(x)$ represents the mutual information between the codeword and x , while $\Phi(\cdot)$ is a function mathematically defined by

$$\Phi(u) = 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}u} e^{-\frac{(y-u/2)^2}{2u}} \log_2(1 + e^{-y}) dy. \quad (11)$$

Then, the mutual information passed from c_i to $v_j \in \tilde{\mathcal{N}}(c_i)$ is computed by

$$\begin{aligned} & I(l_{i \rightarrow j}^{(t)}) \\ &= 1 - \Phi \left[\sum_{j' \in \{j'' \mid v_{j''} \in \tilde{\mathcal{N}}(c_i) \setminus v_j\}} \Phi^{-1} \left(1 - I(l_{j' \rightarrow i}^{(t)}) \right) \right]. \end{aligned} \quad (12)$$

Finally, the mutual information of the soft output from $v_j \in \tilde{\mathcal{N}}(c_i)$ is calculated by

$$I(l_j^{(t)}) \leftarrow \Phi \left[\Phi^{-1} \left(I(l_{i \rightarrow j}^{(t)}) \right) + \Phi^{-1} \left(I(l_{j \rightarrow i}^{(t-1)}) \right) \right]. \quad (13)$$

Meanwhile, we present the process of the layered PEXIT analysis from the perspective of the base graph. Before the decoding begins, $I_{CN, h \rightarrow s}^{(0)}$ is set to 0 for all $h \in \{1, \dots, m\}$, $s \in \{1, \dots, n\}$. At the beginning of the t -th iteration, $I_{APP, s}^{(t)}$ is initialized by $I_{APP, s}^{(t)} \leftarrow I_{APP, s}^{(t-1)}$, $s \in \{1, \dots, n\}$. In the ℓ -th layer, the calculations of the mutual information follow:

1) *VNs to CN:* for all $s \in \{s' \mid b_{\ell, s'} > 0\}$,

$$I_{VN, s \rightarrow \ell}^{(t)} \leftarrow \Phi \left[\Phi^{-1} \left(I_{APP, s}^{(t)} \right) - \Phi^{-1} \left(I_{CN, \ell \rightarrow s}^{(t-1)} \right) \right], \quad (14)$$

2) *CN to VNs:* for all $s \in \{s' \mid b_{\ell, s'} > 0\}$,

$$I_{CN, \ell \rightarrow s}^{(t)} = 1 - \Phi \left\{ \sum_{s' \neq s} \left[\left(1 - \frac{|\mathcal{V}_{s'} \cap \mathcal{S}|}{Z} \right) b_{\ell, s'} \right] \right\}$$

$$\times \Phi^{-1} \left(1 - I_{VN,s' \rightarrow \ell}^{(t)} \right) \Big] + \left[\left(1 - \frac{|\mathcal{V}_s \cap \mathcal{S}|}{Z} \right) \times (b_{\ell,s} - 1) \Phi^{-1} \left(1 - I_{VN,s \rightarrow \ell}^{(t)} \right) \right] \Big\}, \quad (15)$$

3) *VNs Output*: for all $s \in \{s' \mid b_{\ell,s'} > 0\}$,

$$I_{APP,s}^{(t)} \leftarrow \Phi \left[\Phi^{-1} \left(I_{VN,s \rightarrow \ell}^{(t)} \right) + \Phi^{-1} \left(I_{CN,\ell \rightarrow s}^{(t)} \right) \right]. \quad (16)$$

The layered PEXIT analysis proceeds layer-by-layer in correspondence with the layered decoding, completing an iteration upon updating all CN blocks. The following theorem implies that as $Z \rightarrow \infty$, the layered PEXIT analysis can accurately compute the mutual information between the soft outputs of each VN block and the codewords, thereby precisely estimating the decoding performance.

Theorem 2: The mutual information computed by the layered PEXIT analysis from the base graph perspective is identical to that derived based on a tree.

Proof: In the h -th layer of the t -th iteration, v_j passes edge information to c_i and the mutual information is computed by Eq. (10), where $v_j \in \mathcal{N}(c_i)$ and $c_i \in \mathcal{C}_h$. The set of all participating v_j in the update is expressed as $\bigcup_{s \in \{s' \mid b_{h,s'} > 0\}} \mathcal{V}_s$. It is clear that Eq. (10) $\simeq$ Eq. (14) apply, and similarly, Eq. (13) $\simeq$ Eq. (16). In Eq. (12), $\Phi^{-1} \left(1 - I_{j \rightarrow i}^{(t)} \right) = 0$ is guaranteed for all $v_j \in \mathcal{S}$ because $l_{j \rightarrow i}^{(t)} = \infty$. Let $I_{\text{sum}} \triangleq \sum_{j' \in \{j'' \mid v_{j'} \in \mathcal{N}(c_i) \setminus \mathcal{V}_j\}} \Phi^{-1} \left(1 - I_{j' \rightarrow i}^{(t)} \right)$, and we can write

$$\begin{aligned} \mathbb{E}[I_{\text{sum}}] &= \mathbb{E} \left[\sum_{j' \in \{j'' \mid v_{j'} \in \mathcal{N}(c_i) \setminus \mathcal{S}\}} \Phi^{-1} \left(1 - I_{j' \rightarrow i}^{(t)} \right) \right. \\ &\quad \left. - \Phi^{-1} \left(1 - I_{j \rightarrow i}^{(t)} \right) \right] \\ &= \frac{|\mathcal{N}(\mathcal{C}_h) \setminus \mathcal{S}|}{Z} \cdot \mathbb{E}_{v_{j'} \notin \mathcal{S}} \left[\Phi^{-1} \left(1 - I_{j' \rightarrow i}^{(t)} \right) \right] \\ &\quad - \frac{|\mathcal{V}_{[j/Z]} \setminus \mathcal{S}|}{Z} \cdot \mathbb{E}_{v_j \notin \mathcal{S}} \left[\Phi^{-1} \left(1 - I_{j \rightarrow i}^{(t)} \right) \right], \quad (17) \end{aligned}$$

where $|\mathcal{N}(\mathcal{C}_h) \setminus \mathcal{S}|/Z = \sum_{s'} (1 - |\mathcal{V}_s \cap \mathcal{S}|/Z) b_{h,s'}$, and $|\mathcal{V}_{[j/Z]} \setminus \mathcal{S}|/Z = 1 - |\mathcal{V}_{[j/Z]} \cap \mathcal{S}|/Z$. It is also clear that Eq. (12) $\simeq$ Eq. (15). ■

In the BICM systems with $q > 2$, the probability distributions of soft information input to each VN are non-identical. Consequently, the information received by each VN should be modeled as the result of the transmitted bits passing through an equivalent channel and then being demodulated. The mutual information between the demodulated soft information input to the s -th VN block and the transmitted codewords, which is denoted as $I_{APP,s}^{(0)}$, can be obtained by averaging the mutual information between the inputs of each VN within the VN block and the transmitted bit. Reference [42] introduces two calculation methods for bit-channel capacity in BICM systems: error rate to SNR mapping and equivalent channel capacity computation. Within the operational SNR range of BICM systems, these methods yield nearly identical capacity

estimates. The error rate to SNR mapping is adopted in this work for its directness, and $I_{APP,s}^{(0)}$ is initialized by

$$I_{APP,s}^{(0)} = \frac{\sum_{v_g \in \mathcal{V}_s \cap \mathcal{S}} I_g}{Z - |\mathcal{V}_s \cap \mathcal{S}|}, \quad (18)$$

$$I_g = \begin{cases} 0, & v_g \in \mathcal{P} \\ \Phi \left\{ 8 \left[\text{erfc}^{-1} \left(2P_e^{(g')}(\sigma) \right) \right]^2 \right\}, & v_g \notin \mathcal{P} \cap \mathcal{S} \end{cases}$$

$$g' = \left\lceil \frac{\text{pos}(g, \{v_1, \dots, v_{nZ}\} \setminus (\mathcal{S} \cap \mathcal{P}))}{\lceil (nZ - |\mathcal{S}| - |\mathcal{P}|)/q \rceil} \right\rceil, \quad (19)$$

where $\text{pos}(g, \mathcal{A})$ denotes the position index of g in the vector formed by arranging the indices of elements in set $\mathcal{A}$ in ascending order.

Additionally, the assumption of infinite iterations in the traditional PEXIT analysis is invalid for few-iteration decoding. When the iteration counts, denoted as t_M , are small, the convergence judgment of the mutual information $I_{APP,s}^{(t_M)}$ must be modified. In 5G NR systems, the decoding threshold for 5G LDPC codes from the BG1 is defined as the minimum $E_b/N_0 \gamma^*$ at which the PEXIT analysis converges and expressed by

$$\gamma^* = \min \left\{ E_b/N_0 \mid I_{APP,s}^{(t_M)} = 1 - \epsilon, \forall s \in \{1, \dots, 26\} \right\}, \quad (20)$$

where ϵ is a sufficiently small positive scalar. The process of deriving the decoding threshold via PEXIT analysis is defined as $\gamma^* = \Gamma_{L/F}(q, \mathbf{H}_b, Z, \mathcal{P}, \mathcal{S}, t_M)$, where $\Gamma_L(\cdot)$ and $\Gamma_F(\cdot)$ denote the analysis for layered and flooding decoding, respectively.

IV. DESIGN OF MODULATION AND RATE COMPATIBLE PUNCTURING

In this section, we first develop a PMPS algorithm based on PEXIT analysis to design the optimal puncturing for 5G LDPC codes in the BICM systems under specific code rates and modulations. Secondly, according to the MCS tables in the 5G NR standard, we set multiple optimization objectives to be compatible with multiple modulations at a specific code rate. Finally, based on the characteristics of 5G LDPC codes, we integrate PMPS and multi-objective optimization to design the MrcPunc for few-iteration decoding.

A. Progressive Multi-Path Search

In the 5G NR systems, $|\mathcal{P}|$ bits are punctured from the LDPC codes to achieve a target code rate R . Directly optimizing these $|\mathcal{P}|$ positions is unfeasible due to the excessively large search space dimension. Thus, instead of individual bit puncturing, we consider puncturing entire VN blocks. This strategy aligns with [15], simplifies the optimization dimension of the puncturing process to n , and thereby renders the optimization feasible. A vector $\beta = [\beta_i]_{1 \leq i \leq n} \in \{0, 1\}^n$ is defined to describe the puncturing positions, where $\beta_i = 0$ indicates that the i -th VN block of the 5G LDPC codes is not punctured, and $\beta_i = 1$ indicates that the i -th VN block

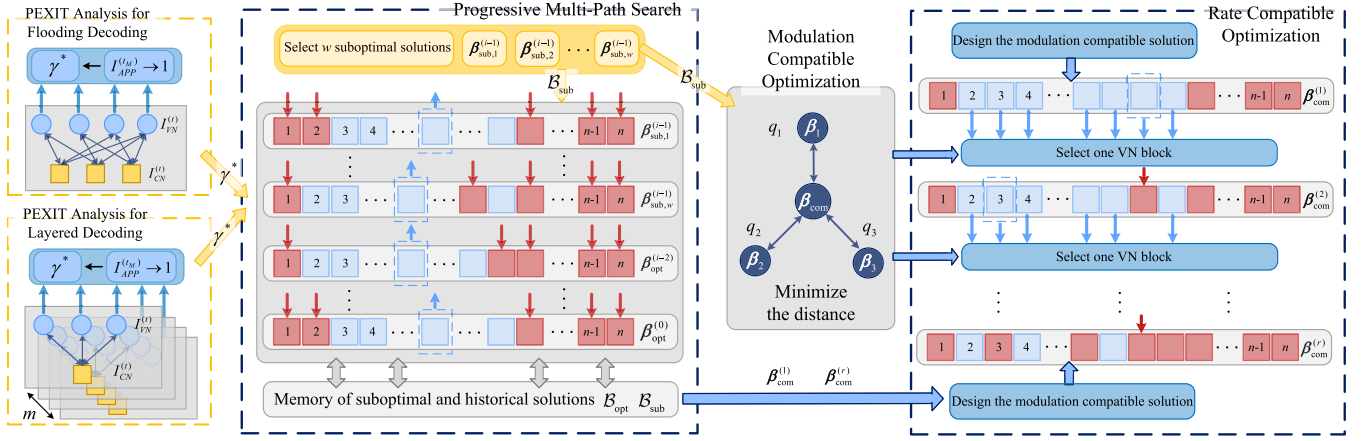


Fig. 4. The design framework of MrcPunc for 5G LDPC codes.

is punctured. The set $\mathcal{P}$ of the punctured VNs, can then be expressed as

$$\begin{aligned} \mathcal{P} &= \mathcal{F}_{\mathcal{P}}(\beta, Z) \\ &= \{v_j \mid \beta_i = 1, (i-1)Z + 1 \leq j \leq iZ\}. \end{aligned} \quad (21)$$

The puncturing can be optimized by minimizing the decoding threshold obtained from PEXIT analysis, and the constraints are that the punctured VN blocks cannot overlap with the VN blocks containing filler bits, and the number of elements with a value of 1 in the puncturing vector β must correspond to the value of $|\mathcal{P}|$. The optimization of the puncturing for 5G LDPC codes with a code rate R using 2^q -QAM in BICM systems can be described as the following optimization problem:

$$\min_{\beta} \Gamma_{L/F}[q, \mathbf{H}_b, Z, \mathcal{F}_{\mathcal{P}}(\beta, Z), \mathcal{S}, t_M] \quad (22a)$$

$$\text{s.t. } \|\beta\|_1 = n - \frac{k/R - |\mathcal{S}|}{Z} \quad (22b)$$

$$\beta_i = 0, i \in \{i' \mid \mathcal{V}_{i'} \cap \mathcal{S} \neq \emptyset\}. \quad (22c)$$

In this optimization problem, there is no VN block punctured partially. If this situation is taken into account, the selection of the fully punctured VN blocks needs to be optimized first, and then the transmitted VN blocks are traversed to find the optimal positions for partial puncturing. In the worst case, the number of traversals is less than n .

If an exhaustive search approach is employed, for the problem (22), at most $\binom{n}{\lceil n/2 \rceil}$ possible solutions need to be explored for a given combination of the code rate and modulation. For instance, for 5G LDPC codes from the BG1, at most 2.85×10^{19} possible solutions require exploration, so it remains challenging to solve the problem by an exhaustive search. In [15], a method called swapping of punctured and transmitted blocks (SPAT) is proposed, which involves pairwise swapping between punctured and transmitted VN blocks. As a greedy algorithm, SPAT can rapidly yield a solution but is prone to getting trapped in local optima. The implementation process of SPAT is depicted in the top-left corner of Fig. 4. In the i -th round of swapping, if swapping the i -th transmitted VN

block with the j -th punctured VN block results in a local optimal solution, then all subsequent swaps starting from the $(i+1)$ -th round will be based on this local optimal solution. Therefore, in this paper, we use PMPS to avoid this situation from occurring as much as possible.

The PMPS conducts each round of swapping based on the set $\mathcal{B}_{\text{opt}}$ of historical optimal solutions and the set $\mathcal{B}_{\text{sub}}$ of the w sub-optimal solutions from the previous round of swapping. Let $\beta_{\text{opt}}^{(i)}$ and $\beta_{\text{sub},j}^{(i)}$ represent the optimal solution and the j -th sub-optimal solution obtained from the i -th round of swapping, respectively. The set of historical optimal solutions is updated in each round, and in the i -th round of swapping, $\mathcal{B}_{\text{opt}}$ includes the optimal solutions obtained from the first to the $(i-1)$ -th rounds of swapping and the puncturing vector $\beta_{\text{opt}}^{(0)}$ ($\mathcal{B}_{\text{opt}} \leftarrow \{\beta_{\text{opt}}^{(0)}, \dots, \beta_{\text{opt}}^{(i-1)}\}$). In the i -th round of swapping, the set $\mathcal{B}_{\text{sub}}$ is composed of the w sub-optimal solutions from the previous round ($\mathcal{B}_{\text{sub}} \leftarrow \{\beta_{\text{sub},1}^{(i-1)}, \dots, \beta_{\text{sub},w}^{(i-1)}\}$). The PMPS, of which the implementation process shown in the bottom-right corner of Fig. 4, explores multiple paths in a single search round and has a backtracking function to prevent the optimization process from getting trapped in local optima. The specific process of PMPS is summarized as Alg. 1, where $\mathcal{T}$ denotes the set of indices for VN blocks that are entirely transmitted in the 5G standard, which ensures that PMPS satisfies the constraint (22c).

B. Modulation Compatible Optimization

In the 5G NR systems, a specific code rate may employ multiple modulations. For example, in Table I, for the LDPC codes with $R = 2/3$, QPSK, 16-QAM, 64-QAM, and 256-QAM are all possible to be employed. The optimal puncturing individually designed for each modulation differs significantly, posing challenges for the implementation of the BICM system corresponding to the MCS tables. There is a need to design the puncturing that achieves good decoding performance across different modulations. Thus, the puncturing optimization involves multiple single objectives, as represented by

$$\min_{\beta} \Gamma_{L/F}[q, \mathbf{H}_b, Z, \mathcal{F}_{\mathcal{P}}(\beta, Z), \mathcal{S}, t_M], q \in \mathcal{Q}, \quad (23)$$

Algorithm 1 The Process of PMPS

Input: $\beta_{\text{opt}}^{(0)}$, w , q , $\mathbf{H}_b$, Z , $\mathcal{T}$, $\mathcal{S}$, and t_M
Output: $\beta_{\text{opt}}^{(|\mathcal{T}|)}$

```

1  $\mathcal{B}_{\text{opt}} \leftarrow \{\beta_{\text{opt}}^{(0)}\}$ ,  $\mathcal{B}_{\text{sub}} \leftarrow \emptyset$ ,  $i \leftarrow 0$ ;
2 for  $s \in \mathcal{T}$  do
3    $i \leftarrow i + 1$ ; // The  $i$ -th round of swapping
4   for  $\beta \in \mathcal{B}_{\text{opt}} \cup \mathcal{B}_{\text{sub}}$  do
5     for  $j \in \{j' \mid \beta_{j'} = 1\}$  do
6        $\beta_j^{(i)} \leftarrow \beta + \mathbf{e}_s - \mathbf{e}_j$ ; // Swap two blocks
7       Determine  $\mathcal{P}_j^{(i)}$  by Eq. (21);
8       Perform PEXIT analysis calculate the
        decoding threshold by Eq. (20);
9     end
10  end
11  Select  $w$  sub-optimal solutions  $\beta_{\text{sub}}^{(i)}$  that minimizes
    the decoding threshold, and update
     $\mathcal{B}_{\text{sub}} \leftarrow \{\beta_{\text{sub},1}^{(i)}, \dots, \beta_{\text{sub},w}^{(i)}\}$ ;
12  Select the optimal solution  $\beta_{\text{opt}}^{(i)}$  that minimizes the
    decoding threshold from the set  $\mathcal{B}_{\text{sub}}$ , and update
     $\mathcal{B}_{\text{opt}} \leftarrow \mathcal{B}_{\text{opt}} \cup \{\beta_{\text{opt}}^{(i)}\}$ ;
13 end

```

where $\mathcal{Q}$ denoting the set of q forming the possible modulation order 2^q . The core objective of problem (23) is to find a solution β that balances performance across various modulations. The Euclidean distance can convert the decoding threshold gaps of candidate solutions under different modulations into a one-dimensional metric for the overall gap. It not only amplifies large gaps to avoid severe performance degradation under any modulation and to ensure the metric sensitivity to overall compatibility, but also facilitates the rapid evaluation of candidate solutions to improve the efficiency of puncturing optimization.

As shown in the middle part of Fig. 4, in the i -th swapping, we first obtain the single-objective solutions $\beta_{\text{opt},q_j}^{(i)}$, $q_j \in \mathcal{Q}$ for each modulation, then derive the modulation compatible solution $\beta_{\text{com}}^{(i)}$ by minimizing the Euclidean distance between the solution vector and these single-objective optimal solutions. The Euclidean distance is evaluated by

$$E(\beta) = \sqrt{\frac{\sum_{q_j \in \mathcal{Q}} \left\{ \Gamma_{L/F} [q_j, \mathbf{H}_b, Z, \mathcal{F}_P(\beta, Z), \mathcal{S}, t_M] \right\}}{-\Gamma_{L/F} [q_j, \mathbf{H}_b, Z, \mathcal{F}_P(\beta_{\text{opt},q_j}^{(i)}, Z), \mathcal{S}, t_M]}}. \quad (24)$$

Therefore, the problem (22) is modified to a multi-objective problem:

$$\begin{aligned} & \min_{\beta} E(\beta) \\ & \text{s.t. Constraints (22b) and (22c).} \end{aligned} \quad (25)$$

The i -th round of swapping in Alg. 1 (line 1 - line 1) involves $|\mathcal{Q}| + 1$ executions, where the first $|\mathcal{Q}|$ times search for single-objective solutions, and the last time optimizes the modulation compatible solution. The i -th round of swapping and the

Algorithm 2 The i -Th Round Swapping and Update of the Sub-Optimal Set in Multi-Objective PMPS

Input: $\mathcal{B}_{\text{opt}}$, $\mathcal{B}_{\text{sub}}$, w , $\mathcal{Q}$, $\mathbf{H}_b$, Z , $\mathcal{S}$, and t_M
Output: $\mathcal{B}_{\text{sub}}$

```

1 for  $q_j \in \mathcal{Q}$  do
2   | Run line 3 - line 10 in Alg. 1;
3 end
4 Select the single-objective solution  $\beta_{\text{opt},q_j}^{(i)}$  for each  $q_j$ ;
5 for  $\beta \in \mathcal{B}_{\text{opt}} \cup \mathcal{B}_{\text{sub}}$  do
6   for  $h \in \{h' \mid \beta_{h'} = 1\}$  do
7      $\beta_h^{(i)} \leftarrow \beta + \mathbf{e}_s - \mathbf{e}_h$ ; // Swap two blocks
8     Evaluate the Euclidean distance by Eq. (24);
9   end
10 end
11 Select  $w$  sub-optimal solutions  $\beta_{\text{sub}}^{(i)}$  that minimizes the
    Euclidean distance;
12  $\mathcal{B}_{\text{sub}} \leftarrow \{\beta_{\text{sub},1}^{(i)}, \dots, \beta_{\text{sub},w}^{(i)}\}$ ;

```

update of the sub-optimal solution set $\mathcal{B}_{\text{sub}}$ in multi-objective PMPS are summarized in Alg. 2.

C. Rate Compatible Optimization

The code rates of 5G LDPC codes vary over a wide range to adapt to changing channel conditions, necessitating that encoders in the 5G NR systems should be compatible with different code rates and the puncturing should align with this requirement. For any two code rates R_1 and R_2 satisfying $R_1 < R_2$, the set of punctured VNs for R_1 , denoted $\mathcal{P}_1$, must be a subset of that for R_2 , denoted $\mathcal{P}_2$. When the code rate R lies in the interval $[R_1, R_2]$, the set $\mathcal{P}$ of punctured VNs for R satisfies $\mathcal{P}_1 \subseteq \mathcal{P} \subseteq \mathcal{P}_2$. Similarly, in the puncturing optimization, for all $R_1 < R_2$, the elements of the corresponding puncturing vectors β_1 , β_2 , and β for rates R_1 , R_2 , and R within the interval $[R_1, R_2]$ satisfy $\beta_i = 1, \forall i \in \{i' \mid \beta_{1,i'} = 1\}$ and $\beta_{2,i} = 1, \forall i \in \{i' \mid \beta_{2,i'} = 1\}$. Thus, the puncturing optimization over this rate interval is described as the problem:

$$\begin{aligned} & \min_{\beta^{(j)}} \Gamma_{L/F} (q, \mathbf{H}_b, Z, \mathcal{F}_P(\beta^{(j)}, Z), \mathcal{S}, t_M), \\ & j \in \{1, \dots, r\}, q \in \mathcal{Q} \end{aligned} \quad (26a)$$

$$\text{s.t. } \|\beta^{(j)}\|_1 = n - \frac{k/R_j - |\mathcal{S}|}{Z} \quad (26b)$$

$$\beta_i^{(j+1)} = 1, \forall i \in \{i' \mid \beta_i^{(j)} = 1\}, j < r \quad (26c)$$

$$\beta_i^{(j)} = 0, i \in \{i' \mid \mathcal{V}_{i'} \cap \mathcal{S} \neq \emptyset\}, \quad (26d)$$

where r represents the number of targeted code rates in puncturing optimization, and R_j represents the j -th code rate among them.

In the 5G standard, different modulations are used for different code rate intervals [29], and the puncturing optimization of 5G LDPC codes needs to refer to the code rate intervals in the MCS tables. Taking 5G LDPC codes from the BG1 as an example, according to Table I, the puncturing optimization needs to be carried out in a total of five code rate intervals.

Algorithm 3 The Nested Puncturing Optimization

Input: r, Q, H_b, Z, S , and t_M
Output: φ

- 1 Obtain $\beta_{\text{com}}^{(1)}$ using Alg. 1 and 2. Reset $\mathcal{T}$ as the input of Alg. 1 and 2 to get $\beta_{\text{com}}^{(r+1)}$. Set $\mathcal{T}_P$ using Eq. (27);
- 2 **for** $i = 1, \dots, r$ **do**
- 3 **for** $q_j \in Q$ **do**
- 4 **for** $h \in \mathcal{T}_P$ **do**
- 5 $\beta_h \leftarrow \beta_{\text{com}}^{(i)} + e_h$;
- 6 Perform PEXIT analysis and calculate the decoding threshold by Eq. (20);
- 7 **end**
- 8 Select β_h that minimizes the decoding threshold, and set $\beta_{\text{com},q_j}^{(i+1)} \leftarrow \beta_h$;
- 9 **end**
- 10 **for** $h \in \mathcal{T}_P$ **do**
- 11 $\beta_h \leftarrow \beta_{\text{com}}^{(i)} + e_h$;
- 12 **end**
- 13 Select β_h that minimizes the Euclidean distance. Set $\beta_{\text{com}}^{(i+1)} \leftarrow \beta_h$ and $\varphi_i \leftarrow h$;
- 14 **end**

To satisfy the constraint (26c) during the design process, we adopt the nested puncturing optimization. For each code rate interval, to ensure that the puncturing corresponding to each code rate in this interval have better performance, we first use the multi-objective PMPS to obtain the modulation compatible puncturing vectors $\beta_{\text{com}}^{(1)}$ and $\beta_{\text{com}}^{(r+1)}$ corresponding to the lowest code rates in the current code rate interval and the next interval, respectively. Then, the set $\mathcal{T}_P$ of indices of VN blocks that are allowed to be punctured is obtained by comparing $\beta_{\text{com}}^{(1)}$ and $\beta_{\text{com}}^{(r+1)}$, and is expressed by

$$\mathcal{T}_P \leftarrow \left\{ i \mid \beta_{\text{com},i}^{(1)} = 0, \beta_{\text{com},i}^{(r+1)} = 1 \right\}. \quad (27)$$

Finally, VN blocks for puncturing are selected from the set $\mathcal{T}_P$ one by one.

When selecting VN blocks, It is still necessary to first search $|Q|$ solutions for each possible modulation in this interval. Then the Euclidean distance is evaluated according to Eq. (24), and the VN block that minimizes the distance after being punctured is selected. For the interval with the highest code rate, It is necessary to first obtain the modulation compatible puncturing vectors corresponding to the lowest and highest code rates in this interval. The nested puncturing optimization is summarized as Alg. 3, the execution of which needs to start from the interval containing the lowest code rate. When designing the puncturing corresponding to the lowest code rate in the next interval, the input $\beta_{\text{opt}}^{(0)}$ in Alg. 1 should satisfy the constraints (26c) and (26d). The output of Alg. 3 is the order of the indices of the VN blocks that are punctured as the code rate increases, which is denoted as $\varphi = [\varphi_i]_{1 \leq i \leq r}$.

V. IMPLEMENTATION OF MODULATION AND RATE COMPATIBLE PUNCTURING

In this section, we first design MrcPunc for 5G LDPC codes from the BG1 under different iteration counts, and then

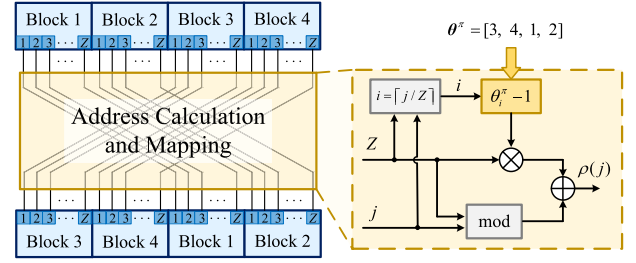


Fig. 5. An example of proposed block interleavers.

propose a block interleaver to enable the practical implementation of MrcPunc. Finally, an approach that combines MrcPunc with IR-HARQ transmission is presented.

A. MrcPunc for 5G LDPC Codes From BG1

The method proposed in Sections IV-A, IV-B, and IV-C is employed to design the MrcPunc for 5G LDPC codes from the BG1 under different iteration counts, and MrcPunc remains consistent across varying numbers $|S|$ of zero-filler bits. The puncturing order φ specifies the sequence of VN blocks to be punctured from lower to higher code rates. The order of elements in φ directly corresponds to the puncturing priority, where the VN blocks with indices positioned earlier in the sequence serve as the priority candidates for puncturing. For 5G LDPC codes from BG1, φ is a vector of length 42 with no duplicate elements. The indices of the punctured VN blocks for the lowest code rate and the puncturing orders for each case are listed in Table II, where “Layered ($t_M = 4$)”, “Layered ($t_M = 8$)”, “Flooding ($t_M = 8$)”, and “Flooding ($t_M = 16$)” represent the LBP decoding with maximum counts of 4 and 8, the FBP decoding with maximum counts of 8 and 16, respectively.

B. Block Interleaver for Implementation of MrcPunc

MrcPunc only modifies the puncturing order of 5G LDPC codes during rate matching. Thus, a block interleaver can be used to prearrange the VN blocks, and the subsequent modules after the block interleaver can remain unchanged as those in the 5G standard. Let $\varphi_1^{(0)}$ and $\varphi_2^{(0)}$ denote the indices of the punctured blocks for the lowest code rate. The reordered sequence of VN blocks is represented by a vector $\theta = [\theta_i]_{1 \leq i \leq 68}$ of length 68 with no duplicate elements, where

$$\theta_i = \begin{cases} \varphi_i^{(0)}, & 1 \leq i \leq 2 \\ \min(\mathcal{F}_i), & 3 \leq i \leq 26 \\ \varphi_{69-i}, & 27 \leq i \leq 68, \end{cases} \quad (28)$$

where $\mathcal{F}_i = \{i' \mid 1 \leq i' \leq 68, i' \notin \{\theta_1, \dots, \theta_{i-1}, \varphi\}\}$. As illustrated in Fig. 5, the block interleaver calculates the memory address for each bit and performs mapping to reorder the VN blocks into the sequence θ . The j -th bit is mapped to the $\rho(j)$ -th bit after interleaving, where $\rho(j)$ is calculated by

$$\rho(j) = (\theta_{\lfloor j/Z \rfloor}^{\pi} - 1)Z + j \mod Z, \quad (29)$$

according to Z and the inverse permutation vector $\theta^{\pi} = [\theta_i^{\pi}]_{1 \leq i \leq 68}$ of θ . Here, θ^{π} satisfies $\forall j \in \{1, \dots, 68\}, \theta_j^{\pi} = i$

TABLE II
THE MRCPUNC FOR 5G LDPC CODES FROM THE BG1

The Indices of Punctured Blocks for $R = 1/3$		The Puncturing Order φ
Layered ($t_M = 4$)	{27, 68}	[35, 28, 63, 42, 45, 67, 65, 48, 59, 66, 62, 56, 58, 60, 64, 46, 47, 57, 50, 61, 51, 52, 55, 39, 40, 54, 53, 49, 2, 36, 41, 43, 44, 33, 32, 38, 30, 37, 29, 34, 31, 5]
Layered ($t_M = 8$)	{2, 27}	[35, 63, 68, 28, 1, 42, 47, 48, 65, 45, 64, 60, 46, 59, 50, 54, 57, 61, 62, 55, 56, 67, 58, 49, 52, 51, 30, 66, 53, 32, 39, 40, 44, 43, 36, 41, 38, 29, 34, 37, 31, 33]
Flooding ($t_M = 8$)	{27, 68}	[42, 28, 35, 63, 48, 45, 65, 66, 67, 59, 62, 56, 60, 61, 50, 47, 46, 1, 40, 54, 2, 32, 30, 58, 51, 55, 64, 57, 39, 53, 38, 36, 49, 44, 52, 43, 33, 34, 37, 31, 29, 41]
Flooding ($t_M = 16$)	{2, 27}	[1, 35, 28, 48, 50, 52, 63, 65, 68, 45, 47, 46, 67, 60, 59, 54, 57, 61, 56, 62, 49, 58, 51, 64, 55, 30, 66, 53, 39, 36, 42, 32, 43, 40, 41, 38, 35, 34, 32, 29, 31, 33]

if and only if $\theta_i = j$. Deinterleaving is executed immediately following puncturing to prevent degradation of the BICM system performance.

In the block interleaver, it is not necessary to perform the complete address calculation in Eq. (29) for each bit. Within the same VN block, if the positions of the first bit before and after interleaving are j and $\rho(j)$ respectively, then the position of the i -th bit in this block after interleaving is $\rho(j) + i$. Thus, the block interleaving incorporates 68Z additions, 68 multiplications, and 68 divisions, all of which are low-bit-width operations. The address calculation in block interleaving can be performed in parallel, and for calculations such as $(\theta_{\lceil j/Z \rceil}^\pi - 1)Z$, the results can also be prestored via a look-up table, resulting in extremely low additional latency and complexity. Such a block interleaver can complete interleaving within a sub- μ s latency, with power consumption only at the milliwatt level [43], [44].

C. Combining MrcPunc With IR-HARQ Transmission

The role of IR-HARQ is to adapt to time-varying channels and provide efficient incremental redundancy transmission for 5G LDPC codes. Except for bits corresponding to the first two VN blocks, all other information bits and parity-check bits are stored in the IR-HARQ buffer. During initial transmission, high-rate 5G LDPC codes are sent starting from the beginning of the buffer. If decoding fails, untransmitted parity-check bits are continuously sent from the buffer; if decoding remains unsuccessful after all bits in the buffer are transmitted, cyclical retransmission is performed from the start. Decoding is performed after combining the soft information from all previous transmissions, thereby improving the reliability [22]. Punctured VNs correspond to untransmitted bits in the buffer, and the vector θ specifies the ordering of bits in the IR-HARQ buffer. When combining MrcPunc with IR-HARQ transmission, the block interleaver and deinterleaver are deployed before and after the IR-HARQ buffer, respectively. At this point, the bits in the IR-HARQ buffer are sorted in a specific order, and the output of the IR-HARQ buffer is directly fed into the BICM system after deinterleaving. Through block interleaving and deinterleaving, MrcPunc is implemented without altering the buffer management, cyclical transmission, and decoding of IR-HARQ transmission, enabling performance gains while retaining the 5G standard procedure.

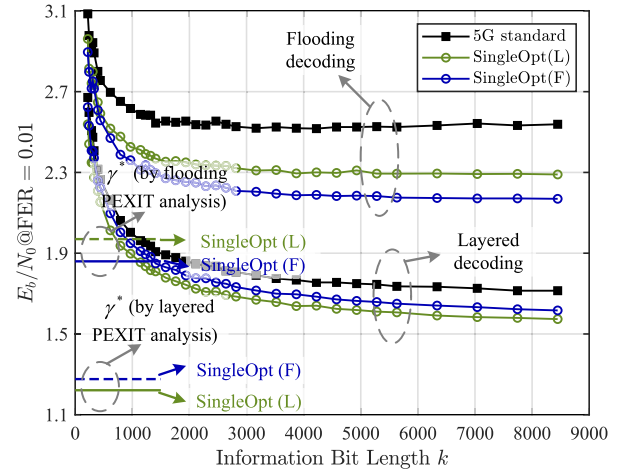


Fig. 6. Performance of different puncturing for 5G LDPC codes from the BG1 with $R = 1/2$ under layered and flooding decoding ($t_M = 8$).

VI. NUMERICAL RESULTS

In this section, we first illustrate the effectiveness of the PMPS algorithm and the accuracy of layered PEXIT analysis. Through extensive simulations, we demonstrate that MrcPunc can significantly improve the performance of few-iteration decoding. Finally, we verify the robustness of the MrcPunc across different iteration counts. In the simulation results, “5G standard” refers to the puncturing specified in [29]. The label “MrcPunc” denotes the puncturing designed in Section V, characterized by its modulation and rate compatibility, whereas “SingleOpt” represents the puncturing optimized for the specific code rate and modulation. These labels are suffixed with “(L)” or “(F)” to indicate optimization based on layered PEXIT analysis (revised) or flooding PEXIT analysis (traditional), respectively. Where necessary, the notation includes “(L/F, $t_M = \dots$)” to specify the maximum iteration counts set during optimization. Additionally, the specific BP decoding schedule (layered or flooding) employed in the simulations is clarified within the corresponding figures and text.

A. Evaluation of PEXIT Analysis and PMPS Algorithm

For intuitive comparison, this paper uses the minimum E_b/N_0 required for the decoder to reach a target frame error rate (FER) of 0.01 as the metric for decoding performance. Fig. 6 illustrates the performance comparison of three types

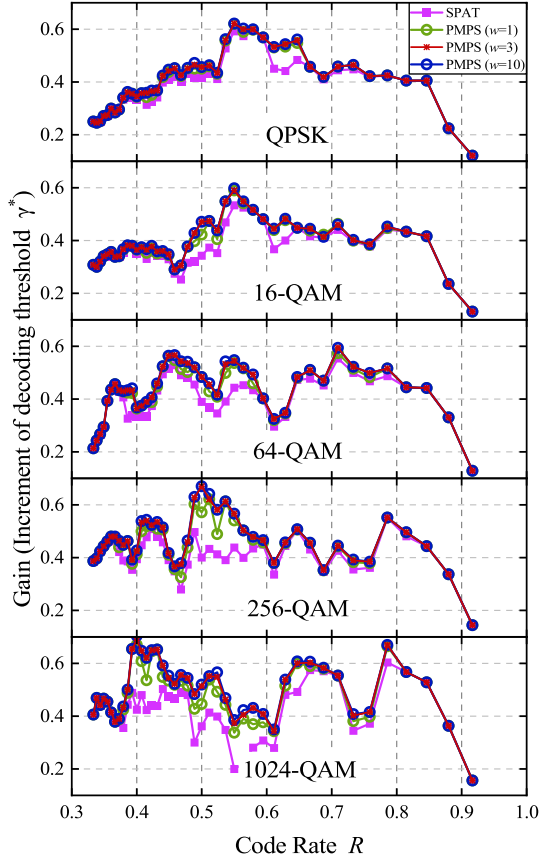


Fig. 7. Gains of puncturing optimized by SPAT and PMPS for the LBP decoding ($t_M = 4$) under different modulations.

of puncturing. As can be observed, the puncturing “SingleOPT (F)” exhibits better decoding threshold in the flooding PEXIT analysis evaluation but poorer performance in layered decoding. Conversely, while the puncturing “SingleOPT (L)” shows better threshold in the layered PEXIT analysis, it performs less effectively in flooding decoding. Notably, each PEXIT analysis provides more accurate evaluation for its corresponding decoding: flooding PEXIT analysis aligns with flooding decoding, and layered PEXIT analysis with layered decoding. This advantage stems from the higher accuracy of tailored PEXIT analysis, underscoring its necessity.

To preliminarily compare the PMPS and SPAT, we optimize the puncturing for each code rate and modulation in the 5G standard using both methods. The gain of each optimization method is defined as the gap between the decoding thresholds obtained via layered PEXIT analysis using puncturing optimized by PMPS with different w (and SPAT) and those of the 5G standard puncturing. Fig. 7 presents the gains from different optimization methods, showing that PMPS achieves larger gains in the intermediate code rate range ($0.4 < R < 0.7$) compared to SPAT, especially as the modulation order increases. To further validate the advantage of the PMPS, we simulate the performance of LBP decoding ($t_M = 4$) for 5G LDPC codes with $R = 1/2$ under 16-QAM, as shown in Fig. 8. Both puncturing optimized by SPAT and PMPS significantly outperform the 5G standard puncturing.

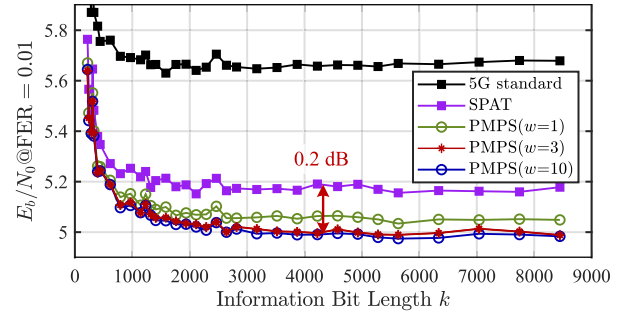


Fig. 8. Performance of the LBP decoding ($t_M = 4$) for 5G-LDPC codes with $R = 1/2$ under 16-QAM.

The puncturing optimized by PMPS with $w = 3$ achieves an additional 0.2 dB gain over that optimized by SPAT. When $w > 3$, the gains from the optimized puncturing become insignificant. The PMPS with $w = 3$ and $w = 10$ yield nearly identical gains despite different puncturing. Given that a larger w introduces extra search complexity, this paper selects $w = 3$ for PMPS to balance complexity and performance.

Although the lifting factor Z is a parameter of the PMPS algorithm, experiments show that it has no obvious impact on the results. For the 5G LDPC codes with $Z \geq 10$, the PMPS algorithm yields identical puncturing. Thus, $Z = 384$ are fixed during optimization in this paper. Considering the worst case, SPAT requires a total of $\lceil \frac{n}{2} \rceil \cdot \lfloor \frac{n}{2} \rfloor$ PEXIT analysis-based decoding threshold calculations for a specific R , while PMPS requires $\left\lceil \frac{(\lceil \frac{n}{2} \rceil + 1) \lfloor \frac{n}{2} \rfloor}{2} \right\rceil + w \left(\lceil \frac{n}{2} \rceil - 1 \right) \cdot \lfloor \frac{n}{2} \rfloor$ calculations. For the 5G LDPC codes from the BG1 and $w = 3$, these respectively correspond to 1156 and 23596 calculations, which are both within acceptable computational complexity.

B. Performance of MrcPunc

Fig. 9 illustrates the performance comparison of LBP decoding with $t_M = 4$ for 5G LDPC codes from the BG1 using different puncturing and modulations. Regardless of the code rate or modulation, the performance of layered decoding after puncturing optimization far exceeds that when using the 5G standard puncturing. The performance of LBP decoding when using the MrcPunc and optimal puncturing is almost the same.

Fig. 10 presents the FER of LBP decoding with $t_M = 4$ versus E_s/N_0 , using different puncturing for 5G LDPC codes with various code rates. The lifting factor Z of these codes is 64, and the transmission adopts QPSK modulation. It is observed that the FER versus E_s/N_0 curve of decoding using MrcPunc exhibits no error floor down to $\text{FER} = 10^{-5}$, with a negligible performance gap compared to decoding using optimal puncturing and a significant gain over that using 5G standard puncturing. Except for the highest code rate $R = 11/12$, decoding using MrcPunc maintains superior performance in the high SNR region relative to that using 5G standard puncturing across all other code rates, verifying the compatibility of MrcPunc under stringent FER requirements.

Fig. 11 presents the performance comparison of the FBP decoding with $t_M = 8$ using different puncturing for 5G LDPC codes with $R = 2/3$. The optimized puncturing still

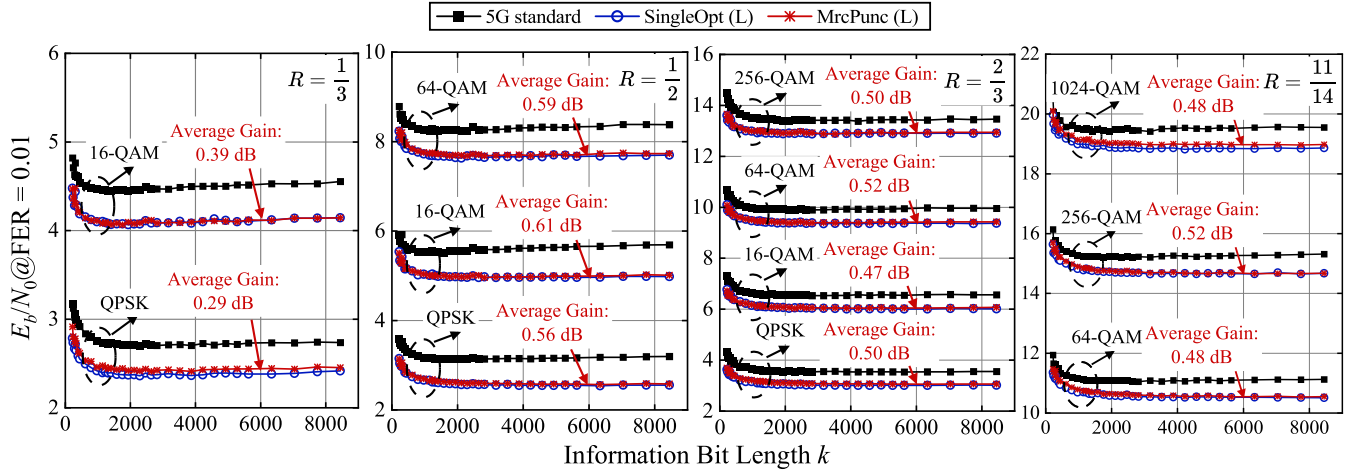


Fig. 9. Performance comparison of the LBP decoding ($t_M = 4$) using different modulations and puncturing for 5G LDPC codes with different code rates.

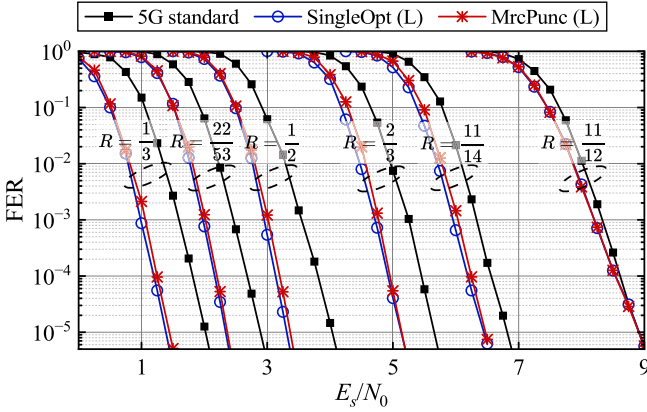


Fig. 10. FER versus E_s/N_0 of the LBP decoding ($t_M = 4$) using different puncturing for 5G LDPC codes with various code rates.

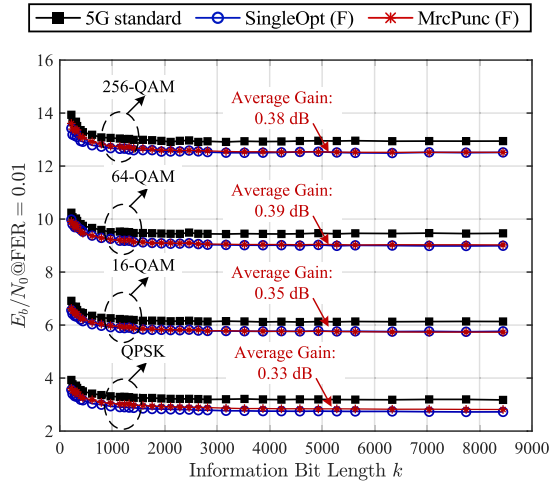


Fig. 11. Performance comparison of the FBP decoding ($t_M = 8$) using different puncturing and modulations.

yields a significant performance improvement, although the gain is slightly reduced to $0.3 \sim 0.4$ dB compared to that in Fig. 9. Figs. 12 and 13 illustrate the performance comparisons of the LBP decoding with $t_M = 8$ and the FBP decoding with $t_M = 16$, using different puncturing for 5G LDPC codes with

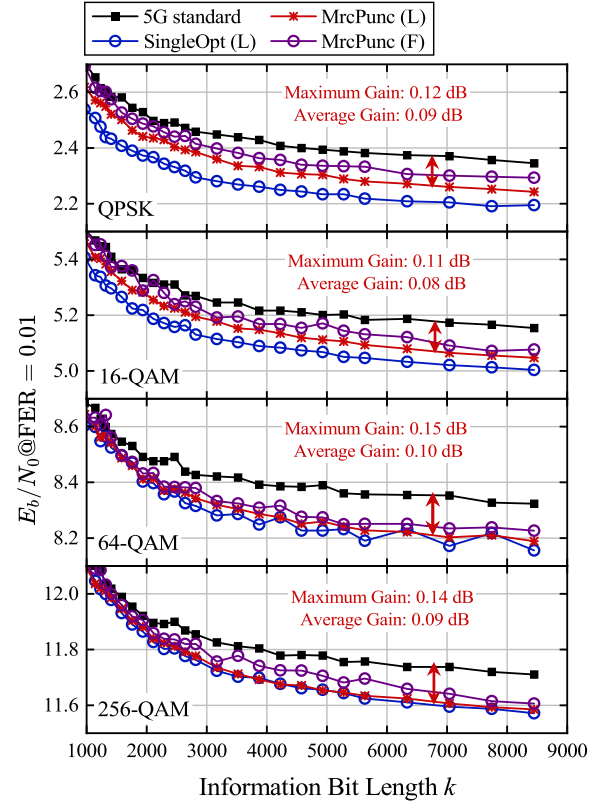


Fig. 12. Performance comparison of the LBP decoding ($t_M = 8$) using different puncturing and modulations.

$R = 2/3$. For both decoding, the performance gains achieved by using either the optimal puncturing for $R = 2/3$ with specific modulations or MrcPunc become marginal. In Fig. 12, we also simulate the performance of LBP decoding using MrcPunc optimized for layered and flooding decoding, where the MrcPunc optimized for layered decoding outperforms that optimized for flooding decoding. This superiority stems from the use of revised PEXIT analysis to evaluate puncturing during optimization, further demonstrating the necessity of the revised PEXIT analysis. Additionally, in Figs. 12 and 13,

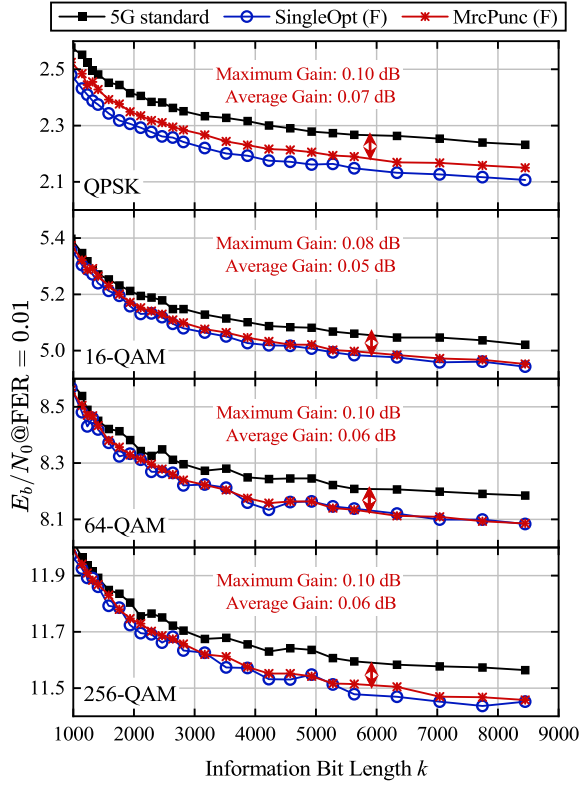


Fig. 13. Performance comparison of the FBP decoding ($t_M = 16$) using different puncturing and modulations.

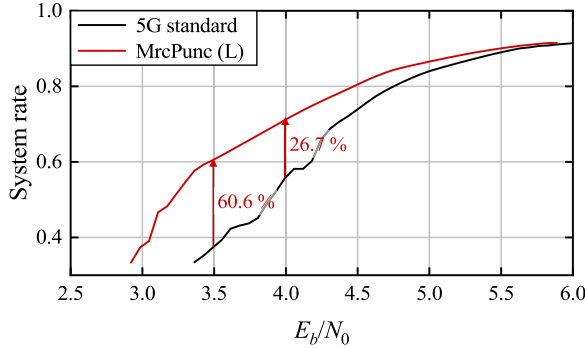


Fig. 14. System rate comparison of LBP decoding ($t_M = 4$) using MrcPunc and 5G standard puncturing under IR-HARQ at target FER of 10^{-5} .

a performance loss is observed when using the MrcPunc compared to the optimal puncturing. In this scenario, given the relatively sufficient iterations, the 5G standard puncturing pattern already enables both decoding to approach their optimal performance, resulting in reduced room for optimization.

C. System Rate Analysis Under IR-HARQ

The system rate is defined as the average code rate of 5G LDPC codes required to meet a target FER, which characterizes the transmission efficiency under IR-HARQ. Fig. 14 presents a system rate comparison of LBP decoding with $t_M = 4$ using MrcPunc versus the 5G standard puncturing, where the target FER is set to 10^{-5} and the lifting factor of 5G LDPC codes is $Z = 64$. It is observed that using

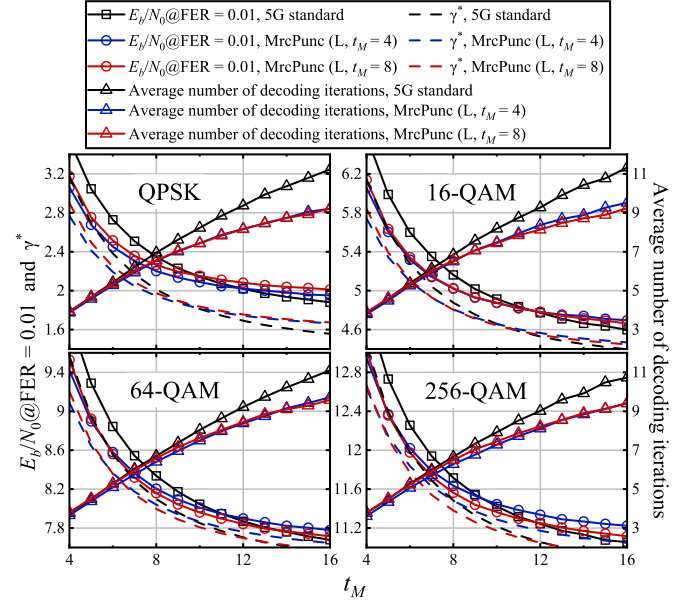


Fig. 15. Comparison of performance, decoding thresholds, and average number of decoding iterations for the LBP decoding with different iteration counts using different puncturing and modulations.

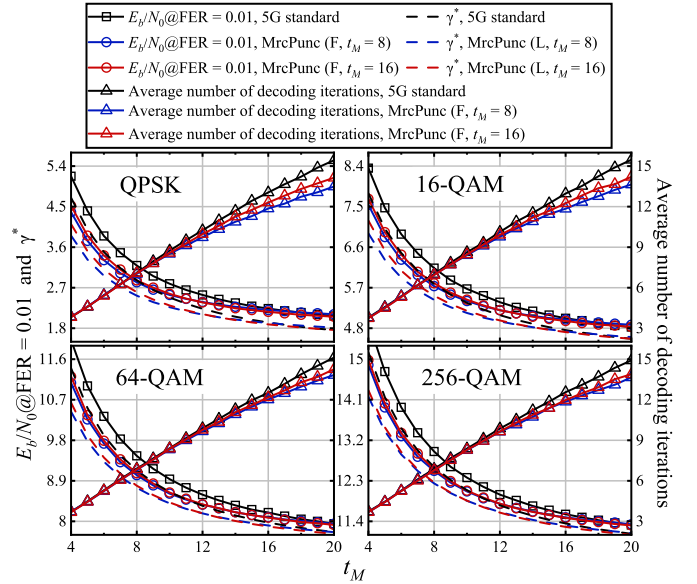


Fig. 16. Comparison of performance, decoding thresholds, and average number of decoding iterations for the FBP decoding with different iteration counts using different puncturing and modulations.

MrcPunc significantly improves the system rate, achieving enhancements of 60.6% and 26.7% at $E_b/N_0 = 3.5$ dB and 4.0 dB respectively, demonstrating its superiority in boosting transmission efficiency under IR-HARQ.

D. Robustness of MrcPunc

Figs. 15 and 16 respectively show the decoding thresholds derived from PEXIT analysis, decoding performance, and average number of iterations of LBP and FBP decoding with different iteration counts, for the 5G LDPC code with $R = 2/3$ and $Z = 384$ using different puncturing. In both Figs. 15

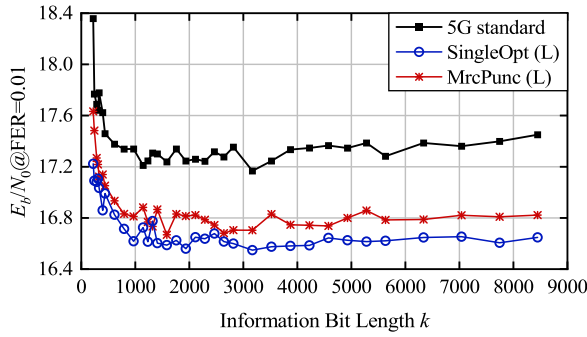


Fig. 17. Performance comparison of the LBP decoding ($t_M = 4$) using different puncturing for 5G LDPC codes with $R = 2/3$ under 1024-QAM.

and 16, dashed lines denote the decoding threshold γ^* obtained from PEXIT analysis, solid triangles represent decoding performance in terms of $E_b/N_0@FER = 0.01$, and solid asterisks indicate the average number of decoding iterations.

It is observed that for both LBP and FBP decoding, as the maximum iteration counts increase, the performance gain of MrcPunc over the 5G standard puncturing gradually diminishes. When the iteration counts are sufficiently large, the 5G standard puncturing becomes optimal, while MrcPunc incurs a slight performance loss. However, in this case, the average number of decoding iterations is relatively reduced, leading to a decrease in complexity. It is also observed that the trend of decoding performance as a function of iteration count is consistent with the trend of decoding threshold changes. This implies that in a system where the decoder information is known to the transmitter, the puncturing that maximizes decoding performance can be selected based on the decoding thresholds to pursue higher transmission reliability.

Fig. 17 presents the performance of LBP decoding ($t_M = 4$) for 5G LDPC codes with $R = 2/3$ under 1024-QAM. Since the combination of $R = 2/3$ and 1024-QAM is not included in Table I, this constitutes a mismatched modulation scenario. The optimal puncturing employed is optimized for the combination of $R = 2/3$ and 1024-QAM. Although MrcPunc incurs a 0.2 dB performance loss compared to the optimal puncturing, it still offers significant performance advantages over the 5G standard puncturing, demonstrating the robustness of MrcPunc under mismatched modulations.

VII. CONCLUSION

This paper proposes MrcPunc to address performance degradation for few-iteration 5G LDPC decoding in URLLC and eMBB. A revised PEXIT analysis, which accounts for the distinct information propagation paths in layered decoding, enables precise puncturing evaluation. By integrating this with the PMPS algorithm and compatibility optimization, MrcPunc is designed to support code rate and modulation combinations in MCS tables while achieving robust decoding performance. Numerical results demonstrate that MrcPunc outperforms 5G standard puncturing, with minor performance loss under larger iteration counts, a trade-off balanced by a reduction in the average number of decoding iterations. Furthermore, MrcPunc guarantees robustness against mismatched modulations and

significantly enhances system rates under IR-HARQ. Future work will extend MrcPunc to other protograph-based LDPC codes, integrate it with specific wireless channel models, and validate its applicability in next-generation mobile networks.

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